

AR Fidget: Augmented Reality Experiences that Support Emotion Regulation through Fidgeting

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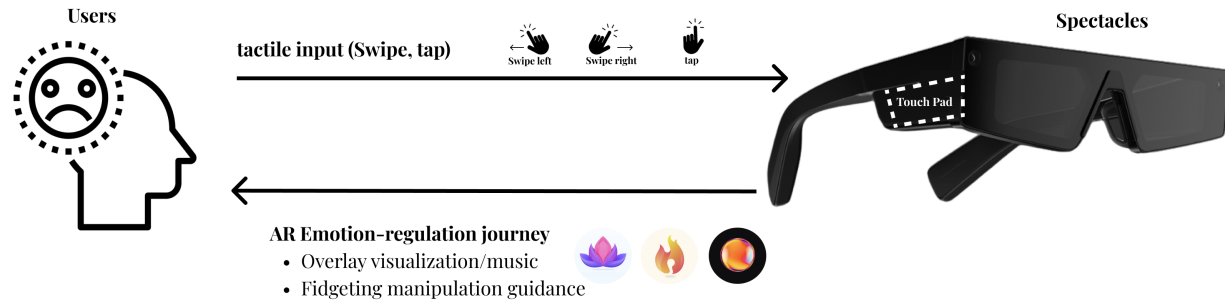


Figure 1: System overview of AR Fidget

ABSTRACT

Emotion regulation—managing one’s emotional state in day-to-day life—is an important set of skills for supporting mental wellbeing. We propose to demonstrate our AR Fidget system that scaffolds emotion regulation using a combination of AR glasses and fidgeting actions. AR glasses have been used in various well-being situations for providing the person with unobtrusive and in-the-moment feedback. Meanwhile, previous studies have shown that the manipulation of soft-bodied ‘fidget’ objects helps people manage emotions and improve focus. This project explores whether combining fidgeting behaviors such as swiping, tapping, and clicking with supportive visual/auditory feedback on AR glasses can support emotion regulation in the moment. We will demonstrate three visual/auditory metaphors in the form of AR lenses (Lotus, Bubble, Fire) combined with suggested target fidget actions based on prior research, toward helping people to regulate anxiety, boredom, and anger.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction devices**.

KEYWORDS

Emotion regulation, Augmented reality (AR), fidgets, haptic input, wellbeing

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1 INTRODUCTION

Emotion Regulation (ER) is the process that an individual monitors, evaluates, and modulates their emotions to accomplish one’s goal [23]. It is a fundamental competence, with effects on life outcomes comparable to IQ or family social status [7, 16]. Research shows these effects are wide-reaching: poorly developed ER leads to increased chances of mental health disorders [1, 13, 14] and societal problems such as criminal behavior, low personal wellbeing [2], and academic under-achievement [6]. Wadley et al. [24] proposed in 2020 that digital emotion regulation was likely to be an important future growth area for providing interventions aimed at assisting emotional self-regulation [7]. Researchers have explored a broad range of research prototypes and commercial products aimed at supporting emotion regulation, such as virtual reality (VR), wearable biosensors, smartphones, or biofeedback techniques [4].

Fidgeting actions such as squeezing, clicking, and stroking involve manipulating objects using repetitive hand movements. According to previous studies [5, 11], fidget actions play an important role in maintaining focus and managing emotions. For example, children are likely to squeeze the fidget objects for releasing angry feelings

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while they prefer to use clicking/pressing/tapping actions when they are bored. Guiding the appropriate target fidget actions based on users' initial undesired states seems to be a promising strategy for regulating emotion.

An important challenge to make emotion regulation useful is that a person in an emotionally challenging situation must remember to implement a regulation strategy while in the moment [21]. Another challenge is that in some situations, it may be best that the regulation strategy employed is subtle and unobtrusive [15]. AR is a promising avenue for providing the person with unobtrusive and in-the-moment feedback that helps improve the affective state. While AR glasses that overlay the interface onto the world have the advantage of allowing users to perceive both reality and augmented information, [18, 20] pointed out that a poorly designed AR interface on the AR glasses would potentially cause distraction. It is important to design the visualization in a minimized way to allow users to sense what is happening to respond to social and environmental cues.

This project explores the potential of combining fidgeting behaviors such as swiping, tapping, and clicking with supportive visual/auditory feedback on AR glasses to support in-the-moment emotion regulation. We focus on two research questions:

- Can fidgeting using the AR glasses touchpad/button (swipe/tap) support emotion regulation?
- Can AR visualization help people move from an undesired emotion state to a desired state?

We demonstrate three visual/auditory metaphors in the form of AR lenses (Lotus, Bubble, Fire) with Snap's Spectacles, toward helping people to regulate anxiety, boredom, and anger. During the AR experiences, users would follow suggested fidget actions (swipe/tap) to interact with visual/auditory metaphors that aim to improve users' affective states. We plan to iterate our prototypes based on the feedback gained from the CHI interactivity session and our university. A more formal user study will be conducted in the future.

2 RELATED WORKS

Since prior research work has collected self-report evidence that both adults and children use fidget objects to aid in emotion regulation [5, 11], researchers started exploring how technologies can augment handheld fidget objects toward more effective support of emotion regulation. BioFidget [12] is a fidget spinner that is integrated with physiological sensing and displays for respiration training. [21] prototypes a hand-crafted plush toy that uses haptic feedback to support in-the-moment calming down strategies for children. While these smart fidgets can provide feedback from the fidget object itself, this is a limited channel for information. Head-mounted displays such as AR and VR seem to be better platforms to give the user feedback that can incorporate imagery and sound feedback that is seen only by the user. In contrast to using most VR headsets, a valuable quality of AR glasses is that they overlay the interface onto the world around the person, allowing them to continue to sense what is happening and to respond to social and environmental cues. Since this advantage overlaps with the goal of

mindful practices that guide people to pay attention to "presence", a number of AR applications have been created for meditation and mindfulness situations when just-in-time feedback and scaffolding is desirable. ZenG [17] combined neurofeedback with EEG with the visual metaphor of Zen Gardening to foster creativity and relaxation with AR display. [19] demonstrates an AR sandbox that intends to support mindfulness practices through visualizing real-time physiological measurements.

Researchers have experimented separately with smart fidgets and with AR visualization applications for supporting emotion regulation, and these works have shown that smart fidgets and interactive AR visualization can add value to emotion regulation strategies. However, to the best of our knowledge, our system is the first that **combines** fidgeting strategies with interactive AR visual/auditory experiences.

3 SYSTEM OVERVIEW

The AR Fidget system uses tactile input and in-the-moment AR visual/auditory feedback to guide people from an undesired emotional state towards a desired state in real-time, all implemented within the Spectacles AR glasses (see Figure 1). The Spectacles controls consist of a swipeable and clickable area on the right glasses temple (side piece of the glasses). These controls allow a user to swipe through a menu, to click to select options, to input fidget actions (swipe/tap) during AR emotion-regulation journey. The system offers three AR experiences for regulating anxiety, boredom, and anger – users can select which one they hope to experience by selecting that option in the menu. The AR interface serves two main purposes: 1) providing the user with guidance about which fidget gestures (swipe and tap) to use in order to help in moving to the desired affective state. 2) providing visualization guidance and accompanying sound that augments the transition from the undesired to the desired state.

4 DESIGN

It is our intent to use a research-through-design process to develop a set of simple, dynamic visualizations (and accompanying audio) that provide ongoing feedback as the user applies the fidget gestures. One of the design goals is to provide engaging visual experiences to guide people in appropriate fidget behavior (swipe and tap) and eventually help users to move from the undesired affective state to a better one. Metaphor is essential to human thought process and communication [25], and visual metaphors are often used in feedback systems for health and wellbeing interfaces (e.g. [25]) as well as in art therapies [3, 22]. Thus the other design goal is to create visual metaphors of affective state trajectories that help users move from the initial affective state to a better state. We described our initial three prototyped visual metaphors in the form of AR lenses that aim to regulate anxiety, boredom, and anger below.

4.1 Lens 1: Lotus (for regulating anxiety)

Buddhist culture uses the lotus as a metaphor for our body and emotion, and some mindful guidance has pointed out that imagining the lotus' opening and closing when breathing and walking helps calm down our body and emotions [8, 9]. Inspired by the concept of

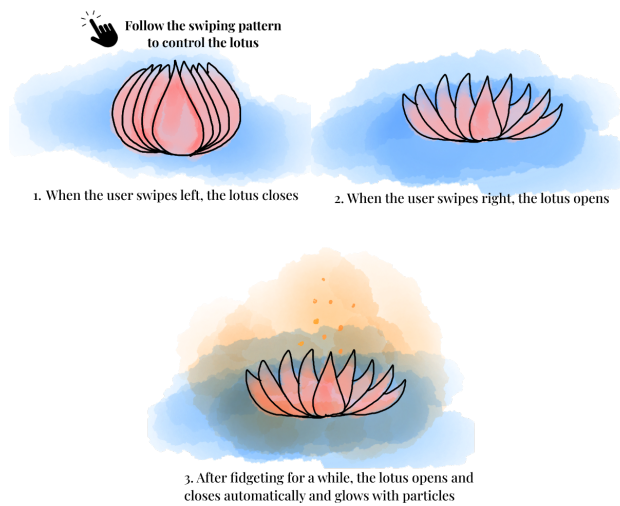


Figure 2: Illustration of AR experiences for regulating anxiety.

the lotus, we designed an interaction in which fidgeting behavior controls the openness and the closedness of the lotus to help shift a user's anxious state toward a calm state. As shown in Figure 2, the AR experience will guide the user to swipe left and right in a regular pattern; if they are following the pattern, the lotus will also close and open with a regular pattern. In the end, the lotus will start to glow, which stands for the calm, positive state the user is moving to.

4.2 Lens 2: Bubble (for regulating boredom)

[5] found that both adults and children will choose tapping behaviors when they feel bored. *Infinite Bubble Wrap* [10] is a never-ending supply of virtual bubble wrap interaction for providing positive affect. Similarly, we hope to integrate tapping behavior into the AR experience by creating a bubble-pop interaction. In our design, there are 5 bubbles in different colors initially; when the user taps the touch bar, random bubble will be popped. The AR journey will provide 4 rounds of "bubble-popping" and after finishing one round, 5 different bubbles will show up again. We hope users move to the calm and alert state after repetitive tapping actions.

4.3 Lens 3: Fire (for regulating anger)

To visualize the initial affective state of "anger", we chose "fire" as a metaphor. [5] pointed out that children prefer squeezing and squishing fidget objects to release anger. Due to the input detection limit of the Spectacles touchpad, instead of squeezing, we offer the user a free-form swiping action as a fidget action to support the release of anger. Users can press as hard as they like (though the Spectacles do not detect relative pressure), and can move as quickly or slowly with their swipes as they like. We anticipate that over the course of the interaction, the user will slow and lighten their pressure, as their anger fades (ending up in a swiping pattern similar to that of the lotus visualization). First, focusing on anger as

Tap the touchbar to pop the bubble

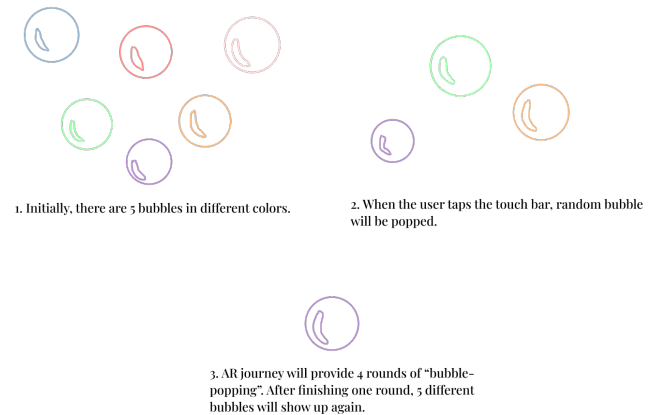


Figure 3: Illustration of AR experiences for regulating boredom.

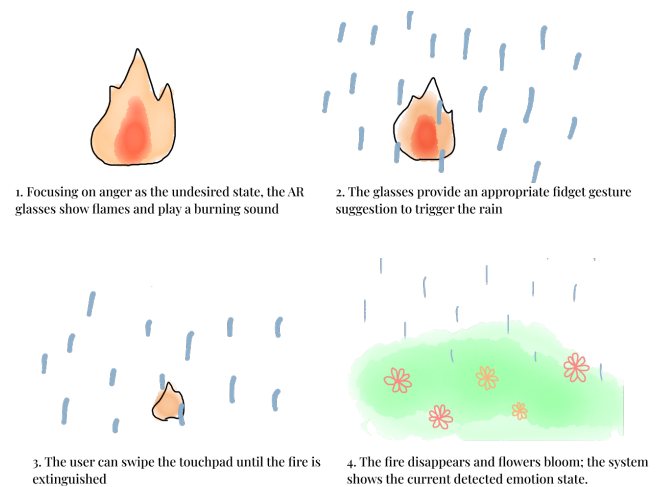


Figure 4: Illustration of AR experiences for regulating anger.

the undesired state, the AR glasses show flames and play a burning sound. We designed the raining effect as the visual feedback for users' swiping actions. After users swipe the touchpad for a while, the rain falls, the flames grow smaller, and eventually, the fire disappears. After the fire has disappeared, the AR glasses show flowers blooming, visualizing the desired better affective state. The following Figure 4 illustrates the walkthrough of this lens.

5 LENS STUDIO PROTOTYPING

We prototyped the interactive AR experiences with Snap's Lens Studio. 2d assets used in the prototype were made in Figma and animated in Adobe After Effect. For the fire effect, a shader graph is used to transform the simple flame texture into the burning fire with a series of noise nodes and color ramp; a "fade" parameter that multiplies the noise was used in the shader graph to control the volume of the fire. The raining effect was implemented with Lens Studio's particle system with a rectangle texture. The lotus

model was made in a 3d software, Blender, and then was imported into the Lens Studio, while the bubble was the default sphere with gradient material. The sound design was done in Logic pro. Since a primary goal of two of the three prototypes is to guide people from an anxious or angry state to a more positive calm state, instruments with gentle and soothing timbre such as harp were used. Audio effects such as reverb and echo were used for creating a warm and calming space. Audio samples like raining, fire burning and bubble-pop sounds were downloaded from freesounds. All the interactions were implemented with Javascript.

6 FUTURE PLANS

We plan to iterate the AR interactions based on informal feedback which we hope to get at CHI 2022 from attendees as well as at our university. A more formal user study will be conducted with a more refined set of prototypes, engaging multiple users with and without the touch interactions to understand the additive value of the touch-based feedback loop. AR Fidget also serves as the initial proof-of-concept validation of the visualization component of our planned just-in-time emotional regulation system that combines a smart fidget (tangible user interface), Augmented Reality glasses, and machine learning to detect patterns of emotional state change related to fidget behavior. The polished version of these AR visualization will be used in that future project.

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